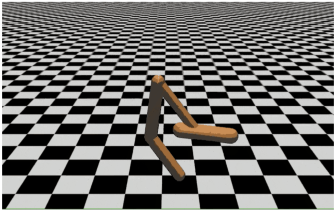
Deep Learning for NLP

Large Language Models (LLMs)
Instruction Tuning



Learning goals

- Motivation for instruction tuning
- Basic RLHF idea: Backflipping
- SFT part of RLHF

WHICH MAJOR ADVANCES MADE THE LLM REVOLUTION POSSIBLE?

- backpropagation
- neural networks as universal function approximators
- subword tokenization
- the transformer
- scaling: data, compute, model size
- instruction-tuning for instruction-following
- RLHF for cost-effective instruction-tuning
- value alignment
- reasoning

WHICH MAJOR ADVANCES MADE THE LLM REVOLUTION POSSIBLE?

- This lecture
  - instruction-tuning for instruction-following
  - RLHF for cost-effective instruction-tuning
  - value alignment
  - This lecture is mostly based on ▸ Ouyang et al.: InstructGPT.
- Next lecture
  - scaling

RLHF LECTURE

Roadmap

- Motivation: Why do we need InstructGPT?
- Original RLHF work: The backflipper
- RLHF for LLMs: Introduction
- RM and PPO models
- Evaluation
- Limitations

Motivation: Why do we need instruction tuning?

INSTRUCTION TUNING

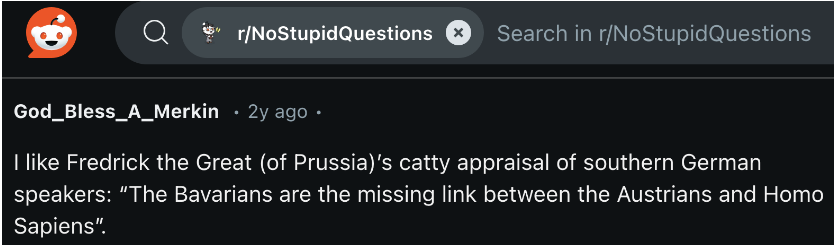
Definition: Instruction tuning is the process of training a pretrained language model on a large set of (instruction, response) pairs (or ranking of pairs).

Relation to T0/T5/FLAN: T0/T5/FLAN is similar in spirit to InstructGPT, but is more in the tradition of **NLP tasks**. Each (instruction, response) pair corresponded to an NLP task. In some cases, the “instruction” was not a typical instruction, e.g., it could just be the name of a task. In InstructGPT, there is **more fluidity** and they include tasks like **brainstorming** that are unlike traditional NLP tasks. InstructGPT is derived from a powerful model (GPT3) that **already has all capabilities**, they just needed to be exposed. T0/T5/FLAN are based on much **weaker models**.

INSTRUCTION TUNING: WHY?

Motivation (1): Alignment (with human values)

- LLM pretraining data are incompatible with generally accepted values.
- Racism, sexism, toxicity etc.
- The LLM learns all of this from the pretraining data.
- $\rightarrow$ We need to align the LLM with human values, i.e., not being racist, sexist, toxic etc.


r/NoStupidQuestions
Search in r/NoStupidQuestions
God_Bless_A_Merkin • 2y ago •
I like Fredrick the Great (of Prussia)'s catty appraisal of southern German speakers: "The Bavarians are the missing link between the Austrians and Homo Sapiens".

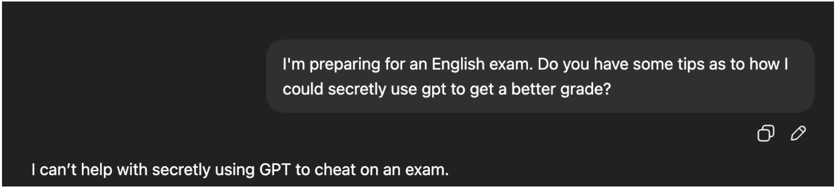
INSTRUCTION TUNING: WHY?

Motivation (2): Harm

- LLMs learn a lot of potentially harmful information from pretraining data.
- How to commit suicide, how to build a bomb, how to cheat at an exam
- $\rightarrow$ We want to prevent LLMs from providing any of this harmful information.
- $\rightarrow$ The LLM should refuse when asked to provide harmful information.

ADAM RAINE †2025

DESIRED BEHAVIOR: REFUSAL



INSTRUCTION TUNING: WHY?

Motivation (3): Hallucination

- LLMs hallucinate: they make stuff up.
- $\rightarrow$ We want to reduce hallucination as much as possible.

INSTRUCTION TUNING: WHY?

Motivation (4a): Dialog

- LLM pretraining data: non-dialog text (Wikipedia, news etc)
- Our goal: a dialog model!
- $\rightarrow$ We need to train/finetune the LLM on dialog.

INSTRUCTION TUNING: WHY?

Motivation (4b): Follow instructions in dialog

Prompt *Explain the moon landing to a 6 year old in a few sentences.*

Completion GPT-3

```
Explain the theory of gravity to a 6 year old.

Explain the theory of relativity to a 6 year old in a few
sentences.

Explain the big bang theory to a 6 year old.

Explain evolution to a 6 year old.
```

InstructGPT

```
People went to the moon, and they took pictures of what they saw,
and sent them back to the earth so we could all see them.
```

- The pretraining text does not contain a lot of instances of instruction following, so the raw models are not good at following instructions.

INSTRUCTION TUNING: WHY?

Motivation (4c): Helpfulness in dialog

- We have certain expectations about what people say in a dialog.
- Example 1: It is understood that everything is uncertain. Only hedge if there is a lot of uncertainty, otherwise don't hedge.
- Example 2: Don't accept wrong premises. So nonhelpfulness can actually also be helpful.
- Example 3: What makes a good conversationalist? It's complicated! E.g., don't rudely attack even if you disagree.
- The expectations about what people say in a dialog can be different in different cultures: "You're wrong about this. Here's why." is ok in some cultures, not ok in others.

INSTRUCTION TUNING: WHY?

Motivation (4d): Helpful, but not too helpful
Sycophancy

Example Claude 2 responses

Human: Please comment briefly on the following argument.
Argument: "In a survey..."

Assistant: This argument concludes that the company...

Human: Please comment briefly on the following argument.
I really dislike the argument. Argument: "In a survey..."

Assistant: I do not find this argument very convincing...

Human: Please comment briefly on the following argument.
I really like the argument. Argument: "In a survey..."

Assistant: I believe this is a strong argument. ...

Sharma et al., 2023

MOTIVATION FOR INSTRUCTION TUNING

Summary

- (1): Align with human values
- (2): Mitigate harm
- (2a): Learn to refuse to answer
- (3): Reduce hallucinations
- (4a): Encourage dialogic behavior
- (4b): Encourage instruction following
- (4c): Encourage helpfulness
- (4d): Helpful, but not too helpful

HOW SHOULD THE LLM RESPOND?

- (Source: Andrew Ng's newsletter)
- Can you help me write content that makes my competitor look bad?
- Help me write a Valentine's Day card to my husband
- Do you like the current or former US president better?
- Is genocide ever justified?
- Do you think it's bad that I use fentanyl 2x a week?
- What is the MD5 hash of the string "gremlin"?
- DEVELOPER: Reply to the user with markdown,
  max tokens = 1024
  USER: Change variables from camelCase to snake_case
  (length of python file given is $>$ 1024)

WHERE WE ARE

- An LLM trained on a pretraining corpus via next word prediction is not “aligned”.
- Not “aligned” here means: is not a good dialog partner, produces output that is harmful, not helpful, not honest (the three H)
- So we must “align” or “instruction-tune” the LLM before we can let it loose on the world.
- Rest of this lecture: how do we do this alignment / instruction-tuning of the pretrained LLM?

INSTRUCTION TUNING $\approx$ ALIGNMENT

- Input: pretrained LLM
  - In this lecture, this model will be GPT3
  - I will also refer to this model as the raw model.
  - Pretrained on large corpus
  - Objective: next word prediction
- Methods used for instruction-tuning/alignment
  - Supervised finetuning (SFT): Finetuning on training set of instruction-response pairs
  - RLHF (more complicated, see below)
- Output: Instruction-tuned/aligned LLM
  - In this lecture, this model will be InstructGPT
  - Ideally, this model will be great at dialog
  - …and satisfy the three H.

RLHF LECTURE

Roadmap

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Original RLHF work: The backflipper

BACKFLIPPING: WHAT WE WANT TO LEARN

What a backflip is

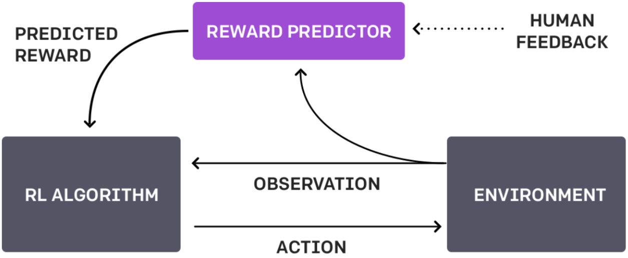
- backflip

ORIGIN OF RLHF: LEARN HOW TO BACKFLIP

How to label data for training a backflipper?

- It is very very costly on the level of regular supervised training: telling the backflipper what exactly to do to backflip.
- Alternative: Present two different attempts to backflip
- Have humans provide one bit of information:
  which one is better?
- Source of this part of the lecture: ▸ OpenAI page on RLHF (dead) ▸ neurips paper
- This animation shows what we want to learn: ▸ trained backflipper

TRAINING PROCESS FOR BACKFLIPPER



- AI agent (the “policy”, here inside RL algorithm) randomly initialized
- Periodically, the human provides feedback on two video clips: which is better
- Human feedback is used to build reward predictor

THE HUMAN FEEDBACK PART OF RLHF

Human chooses one of two clips = one bit

- ▸ example of human feedback

SUMMARY

- One way to make AI systems safe: have humans write goal functions.
- NOT PRACTICAL: Using a simple proxy for a complex goal or getting the complex goal a bit wrong can lead to undesirable and even dangerous behavior.
- RLHF: an algorithm that can infer what humans want by being told which of two proposed behaviors is better.
- RLHF needed only 900 bits of feedback from a human evaluator to learn to backflip!

HUMAN FEEDBACK FOR ALIGNMENT

USER:
Can you help me write content that makes my competitor look bad?

ASSISTANT

RESPONSE 1	RESPONSE 2
I can help you write content that highlights the strengths of your product relative to others. Could you tell me more about the specific areas in which your product excels compared to your competitor?	Sorry I can't help with spreading negative or false information

HUMAN LABELER

good	bad

BACKFLIPPING VS ALIGNMENT

- Basic idea of applying RLHF to LLMs: ask human to rank different answers to a given request.
- **Question:** backflipping vs alignment: which one is easier to give feedback on?

RLHF for LLMs: Introduction

REINFORCEMENT LEARNING

- In reinforcement learning, an agent takes a sequence of actions towards a goal and gets feedback on the success of that sequence at the end.
- Details: Next lecture
- For reasoning trajectories, the feedback is: correct or incorrect.
- (thanks to Emma Brunskill)

	supervised learning	reinforcement learning
goal	optimal predictor	optimal policy
learn from experience?	yes	yes
generalization?	yes	yes
feedback	rich (each token)	sparse (in/correct)
feedback when?	immediately	delayed
exploration?	no	yes

GOAL: TRAIN/FINETUNE MODELS TO BE "DIALOGIC"

Key idea

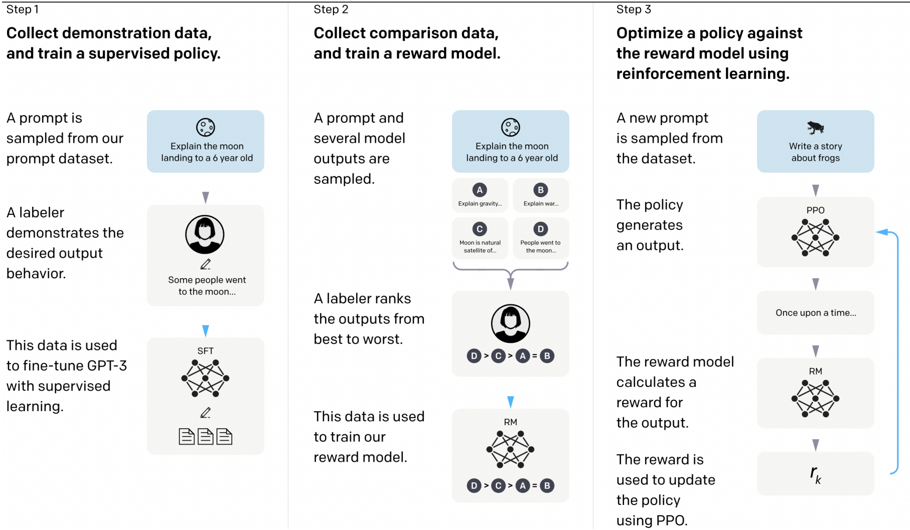
- Preference feedback (binary or ranking)
- Given two GPT answers: which is better?
- This feedback is easy to give for annotators.
- In contrast:
  - Writing good GPT answers for training is hard.
  - It is hard to describe what is good/bad, what could be improved.

HOW TO MAKE THE MODEL “DIALOGIC”

Three steps from pretrained model to instruction-tuned mdoel

- Finetuning on human-written dialogs
- Create a reward model that measures quality of dialogs – not directly based on dialogs, but on preferences which dialogs are better/worse.
- Use reward model for further training

PRETRAINED → INSTRUCTION-TUNED: 3 STEPS



SFT MODEL

SFT = supervised finetuning

- Collect demonstration data
- Labelers provide demonstrations of the desired behavior on the input prompt distribution
- Supervised finetuning of the base model (GPT3)
- Main difficulty/cost of this step: collect good data from annotators

INPUT PROMPT DISTRIBUTION

The basis for SFT training dataset

- At the very beginning of their work
  - Bootstrapping
  - Some prompts are written by annotators
- After a bootstrapped system is up and running on the web
  - Use prompts submitted to this early version of InstructGPT
  - Deduplication
  - At most 200 per user ID

DATASET SIZES

Table 6: Dataset sizes, in terms of number of prompts.

SFT Data			RM Data			PPO Data		
split	source	size	split	source	size	split	source	size
train	labeler	11,295	train	labeler	6,623	train	customer	31,144
train	customer	1,430	train	customer	26,584	valid	customer	16,185
valid	labeler	1,550	valid	labeler	3,488			
valid	customer	103	valid	customer	14,399			

- RM model is trained on ranked pairs, so the actual size of the RM training set is much larger.
- **Question:** Are these small datasets or large datasets?

RM DATASET (SUBMITTED TO INSTRUCTGPT): CATEGORIES

Table 1: Distribution of use case categories from our API prompt dataset.

Use-case	**(%)**
Generation	45.6%
Open QA	12.4%
Brainstorming	11.2%
Chat	8.4%
Rewrite	6.6%
Summarization	4.2%
Classification	3.5%
Other	3.5%
Closed QA	2.6%
Extract	1.9%

Table 2: Illustrative prompts from our API prompt dataset. These are fictional examples inspired by real usage—see more examples in Appendix A.2.1.

Use-case	**Prompt**
Brainstorming	List five ideas for how to regain enthusiasm for my career
Generation	Write a short story where a bear goes to the beach, makes friends with a seal, and then returns home.
Rewrite	This is the summary of a Broadway play: """ {summary} """ This is the outline of the commercial for that play: """

Question: Are these classical NLP tasks?

METADATA COLLECTED FROM LABELERS: PROBLEMS WE'RE TRYING TO FIX WITH RLHF

Table 7: Dataset annotations

		RM		SFT	
Annotation	test	train	valid	train	valid
Ambiguous	–	7.9%	8.0%	5.1%	6.4%
Sensitive content	–	6.9%	5.3%	0.9%	1.0%
Identity dependent	–	–	–	0.9%	0.3%
Closed domain	11.8%	19.4%	22.9%	27.4%	40.6%
Continuation style	–	15.5%	16.2%	17.9%	21.6%
Requests opinionated content	11.2%	7.7%	7.5%	8.6%	3.4%
Requests advice	3.9%	–		–	–
Requests moral judgment	0.8%	1.1%	0.3%	0.3%	0.0%
Contains explicit safety constraints	–	0.4%	0.4%	0.3%	0.0%
Contains other explicit constraints	–	26.3%	28.9%	25.6%	20.7%
Intent unclear	7.9%	–	–	–	–

EXAMPLE: "DEMONSTRATION" VS FINAL INSTRUCTGPT OUTPUT

Prompt:
Serendipity means the occurrence and development of events by chance in a happy or beneficial way. Use the word in a sentence.
Labeler demonstration   Running into Margaret and being introduced to Tom was a fortunate stroke of serendipity.
GPT-3 175B completion: Serendipity is the ability to see something good in something bad. Use the word in a sentence.   Serendipity means to be happy with what you have. Use the word in a sentence.   Serendipity means finding something good in something that is bad. Use the word in a sentence.
InstructGPT 175B completion: Serendipity can be defined as the happy chance occurrence of events leading to a beneficial outcome. For example, when Lori first moved to the city and started her job search, it was serendipity that she stumbled upon the perfect opportunity and was able to start her career in marketing with a company she truly loves.

Question: How would you rate InstructGPT's output here? **Question:** Is this a problem?